

Check-In 08/20. (True/False) Every linear function must have a constant rate of change.

Solution. The statement is *true*. This is one of the equivalent conditions for a function $f(x)$ to be linear, along with $f(x)$ having the form $f(x) = mx + b$ and $f(x)$ having a graph that is a line. In fact, we can see that a linear function has a constant average rate of change. Given any distinct values x_1, x_2 and linear function $f(x) = mx + b$, the average rate of change on the interval containing x_1, x_2 as endpoints is...

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{(mx_2 + b) - (mx_1 + b)}{x_2 - x_1} = \frac{mx_2 + b - mx_1 - b}{x_2 - x_1} = \frac{mx_2 - mx_1}{x_2 - x_1} = \frac{m(x_2 - x_1)}{x_2 - x_1} = m$$

So, the average rate is constant and equal to the slope of $f(x)$ — m .

Check-In 08/24. (True/False) If $f(x) = x + 3$ and $g(x) = 5$, then $(f \circ g)(0) = 15$.

Solution. The statement is *false*. Observe that $(f \circ g)(0) = f(g(0)) = f(5) = 5 + 3 = 8$. Recall that the notation ‘ $\circ$ ’ is function composition—not function multiplication. The mistake here is that ‘ $\circ$ ’ is treated as multiplication: $(f \circ g)(0) \neq (fg)(0) = f(0)g(0) = (0 + 3) \cdot 5 = 3 \cdot 5 = 15$.

Check-In 08/25. (True/False) The function $f(x) = 3(x + 5)^2 - 6$ is quadratic with vertex $(5, -6)$ and has a maximum possible output.

Solution. The statement is *false*. The function $f(x)$ is clearly quadratic, as expanding yields $f(x) = 3x^2 + 30x + 69$, which is quadratic. Alternatively, we see that $f(x)$ has the vertex form of a quadratic function, $a(x - P)^2 + Q$, where a is the a from the standard form of a quadratic $ax^2 + bx + c$ and (P, Q) is the vertex of the quadratic function. Observe $f(x) = 3(x + 5)^2 - 6 = 3(x - (-5))^2 + (-6)$ is in vertex form with $a = 3$, $P = -5$, and $Q = -6$. Therefore, the vertex is $(-5, -6)$ —not $(5, -6)$. Moreover, because $a = 3 > 0$, the quadratic function opens upwards so that it has a minimum possible output (namely, the y -coordinate of the vertex, -6) but no maximum possible output.

Check-In 08/26. (True/False) A quadratic function $f(x)$ has a y -intercept, two x -intercepts, and either a maximum or a minimum output.

Solution. The statement is *false*. We know that a quadratic function is defined everywhere. So, $f(0)$ is defined. But this is precisely the y -intercept for $f(x)$. We know that each quadratic function has either a maximum value (if $a < 0$ so that the quadratic function opens downwards) or a minimum value (if $a > 0$ so that the quadratic function opens upwards). However, not all quadratic functions have two x -intercepts. By the Fundamental Theorem of Algebra, a quadratic (polynomial of degree 2) has either none, one, or two x -intercepts. For instance, $f(x) = x^2 + 1$ has no x -intercept (simply choose any quadratic with a vertex in Quadrant I or Quadrant II that opens upwards— $a > 0$ —or any quadratic with a vertex in Quadrant III or Quadrant IV that opens downwards— $a < 0$), $f(x) = x^2$ has a single x -intercept ($x = 0$), and $f(x) = (x - a)(x - b)$ has two x -intercepts ($x = a, b$)—so long as $a \neq b$, e.g. $f(x) = (x - 1)(x + 1) = x^2 - 1$.

Check-In 08/27. (*True/False*) If the discriminant of a quadratic function $q(x)$ is -16 , then $q(x)$ does not factor over the real numbers but does factor ‘nicely’ over the complex numbers.

Solution. The statement is *true*. We know that a quadratic function $ax^2 + bx + c$ factors over the real numbers if and only if the discriminant, $D = b^2 - 4ac$, is nonnegative, i.e. $D \geq 0$. So, because the given discriminant is negative, we know that $q(x)$ does not factor over the real numbers. We know also a quadratic function only factors ‘nicely’ (over the real numbers or complex numbers) if $|D|$ is a perfect square. We know that $|-16| = 16 = 4^2$, so that the discriminant is a perfect square. Therefore, $q(x)$ does not factor over the real numbers but does factor ‘nicely’ over the complex numbers. For instance, $q(x) = x^2 - 6x + 13$ has discriminant -16 and does not factor over the real numbers, but does factor over the complex numbers as. . .

$$x^2 - 6x + 13 = (x - (3 + 2i))(x - (3 - 2i))$$

In fact, every quadratic function with discriminant -16 has the form $ax^2 + bx + \frac{b^2 + 16}{4a}$, where a, b are real numbers and $a \neq 0$. [Try proving this is the case using quadratic formula and the Fundamental Theorem of Algebra.] Our previous example chose $a = 1$ and $b = -6$.